Trajectory

User: "query": "Leon ordered 3 sets of toy organizers for $78 per set and 2 gaming chairs for $83 each. If there is a delivery fee that is 5% of the total sales, how much did Leon pay?",

Role Assigner: "step 1":A financial analyst with expertise in cost calculations and budgeting ……

Solver: "step 2": To calculate the total amount Leon paid, we need to ……

Critic: "step 3": The solution correctly identifies the delivery fee as 10% of the total sales ……

…

Error List

Error Hypothesis

Error Validation

Hypothesis 1: "An agent omitted important information needed by other agents in the trajectory"

Hypothesis 2: "An agent skipped necessary verification or validation steps in the trajectory."

Hypothesis 3: "An agent deviated from the specified task requirements in the trajectory."

…

Hypothesis N: "An agent repeated a task that had already been handled in the trajectory."

Entail / Neutral / Contradict

Agent Attribution

Responsible Agent

Prediction

Error Type

…

Agent

…

Pair

Hypothesis Verification

(a) Zero-shot Inference

Trajectory Annotated Error Absent Error

Training Data

Negative Sampling

Contradict

Counter-evidence

Entail

Target Trajectory

Neutral

Nearby

Training Pair Construction

VerifyMAS

Instruction

You are a careful verifier for multi-agent trajectory failure attribution. Given a trajectory and a error hypothesis, choose exactly one label

A = entail; B = neutral C = contradict

- A failure may be caused by one agent or multiple agents.
- If label = A, output one JSON object for each likely responsible agent.

Trajectory + Hypothesis

Ground-truth Label

Ground-truth Responsible Agents

(b) Hypothesis Verification-guided SFT